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Semester	B.E. Semester VII - CMPN
Subject	R Programming Lab
Student Name	Prasana Shangloo
Roll Number	23102A0076
Branch	CMPN A
Grade and Subject Teacher's Signature	

Experiment Number	**5**
Experiment Description	**Lab 1: Statistical Analysis of Palmer Penguins: Descriptive Statistics & Hypothesis Testing** Topic: Descriptive Statistics, Hypothesis Testing | Dataset: Palmer Penguins Dataset (UCI Machine Learning Repository) **Problem Statement** Researchers studying penguins in the Palmer Archipelago, Antarctica, collected measurements from three species (Adelie, Chinstrap, Gentoo) to understand variation in physical characteristics. This lab investigates whether body mass differs significantly across species and sex, combining descriptive statistics, hypothesis testing and effect-size estimation. After removing rows with missing values, 333 of the original 344 penguin records were used.

Process	**Task 1: Descriptive Statistical Analysis of Body Mass**
	Overall and species-wise descriptive statistics were computed for body_mass_g, along with a histogram, species-wise boxplot and density plot.
	`desc_stats <- function(x) {`
	`  c(`
	`  Mean = mean(x),`
	`  Median = median(x),`
	`  Min = min(x),`
	`  Max = max(x),`
	`  Variance = var(x),`
	`  SD = sd(x),`
	`  Q1 = as.numeric(quantile(x, 0.25)),`
	`  Q3 = as.numeric(quantile(x, 0.75)),`
	`  IQR = IQR(x),`
	`  Skewness = skewness(x),`
	`  Kurtosis = kurtosis(x)`
	`  )`

```
}

overall_desc <- desc_stats(penguins$body_mass_g)
round(overall_desc, 2)

     Mean    Median       Min       Max  Variance        SD        Q1
Q3
  4207.06   4050.00   2700.00   6300.00 648372.49    805.22   3550.00
4775.00
      IQR  Skewness  Kurtosis
  1225.00      0.47      2.26
```

Species-wise statistics (dplyr summary and psych::describeBy). Note: the tibble printer truncated the Kurtosis column due to console width: the describeBy() output below it (from the same run) reports per-species kurtosis instead.

```
species_desc <- penguins %>%
  group_by(species) %>%
  summarise(
    Mean     = mean(body_mass_g),
    Median   = median(body_mass_g),
    Min      = min(body_mass_g),
    Max      = max(body_mass_g),
    Variance = var(body_mass_g),
    SD       = sd(body_mass_g),
    Q1       = quantile(body_mass_g, 0.25),
    Q3       = quantile(body_mass_g, 0.75),
    IQR      = IQR(body_mass_g),
    Skewness = skewness(body_mass_g),
    Kurtosis = kurtosis(body_mass_g)
  )
print(species_desc)

describeBy(penguins$body_mass_g, group = penguins$species)

# A tibble: 3 × 12
  species    Mean Median   Min   Max Variance    SD    Q1    Q3   IQR
Skewness
  <fct>     <dbl>  <dbl> <int> <int>    <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl>
1 Adelie    3706.   3700  2850  4775  210332.  459. 3362.  4000  638.
0.277
2 Chinstrap 3733.   3700  2700  4800  147713.  384. 3488.  3950  462.
0.242
3 Gentoo    5092.   5050  3950  6300  251478.  501. 4700   5500  800
0.0450
# i 1 more variable: Kurtosis <dbl>
 Descriptive statistics by group
group: Adelie
   vars   n    mean     sd median trimmed    mad  min  max range skew
kurtosis
X1    1 146 3706.16 458.62   3700 3691.53 444.78 2850 4775  1925 0.27
-0.62
      se
X1 37.96
------------------------------------------------------------
group: Chinstrap
   vars  n    mean     sd median trimmed    mad  min  max range skew
kurtosis
X1    1 68 3733.09 384.34   3700 3719.64 370.65 2700 4800  2100 0.24
0.36
      se
X1 46.61
------------------------------------------------------------
group: Gentoo
   vars   n    mean     sd median trimmed    mad  min  max range skew
kurtosis
X1    1 119 5092.44 501.48   5050 5090.46 593.04 3950 6300  2350 0.04
-0.79
```

```
      se
X1 45.97
```

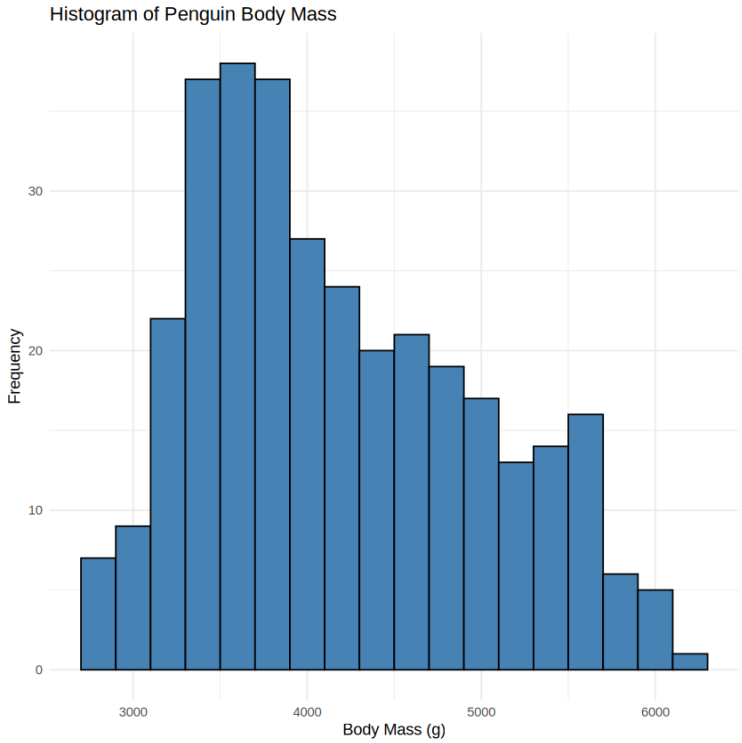


Figure 1.1: Histogram of penguin body mass

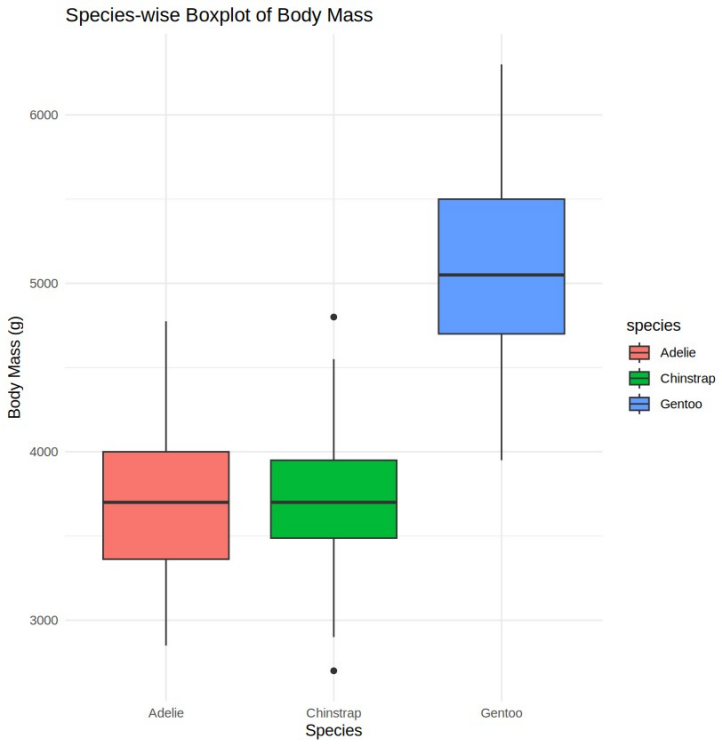


Figure 1.2: Species-wise boxplot of body mass

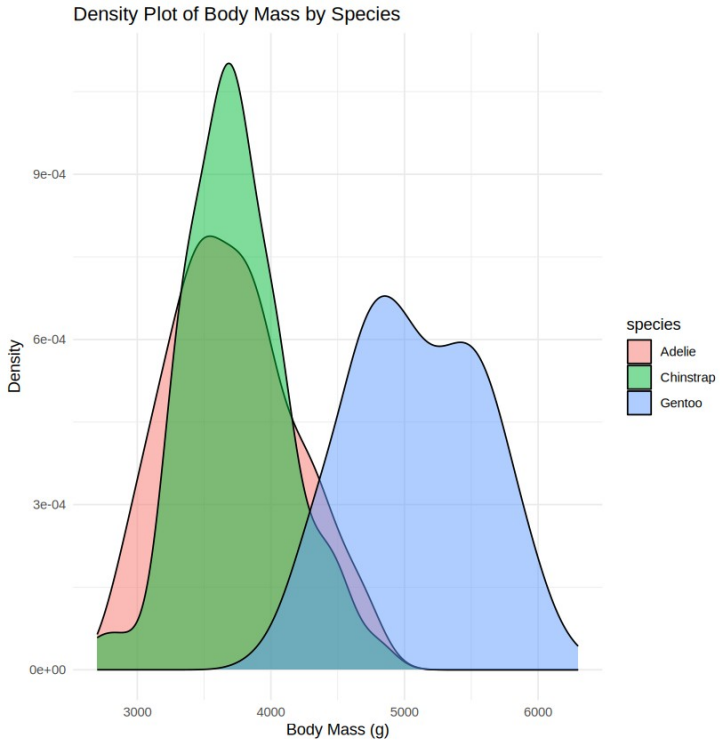


Figure 1.3: Density plot of body mass by species

Task 2: Hypothesis Testing (Male vs Female Body Mass)

H0: mean body mass is equal for males and females. H1: mean body mass differs by sex.

```
shapiro.test(male_mass)
shapiro.test(female_mass)

       Shapiro-Wilk normality test

data:  male_mass
W = 0.92504, p-value = 1.227e-07

       Shapiro-Wilk normality test

data:  female_mass
W = 0.91931, p-value = 6.155e-08
```

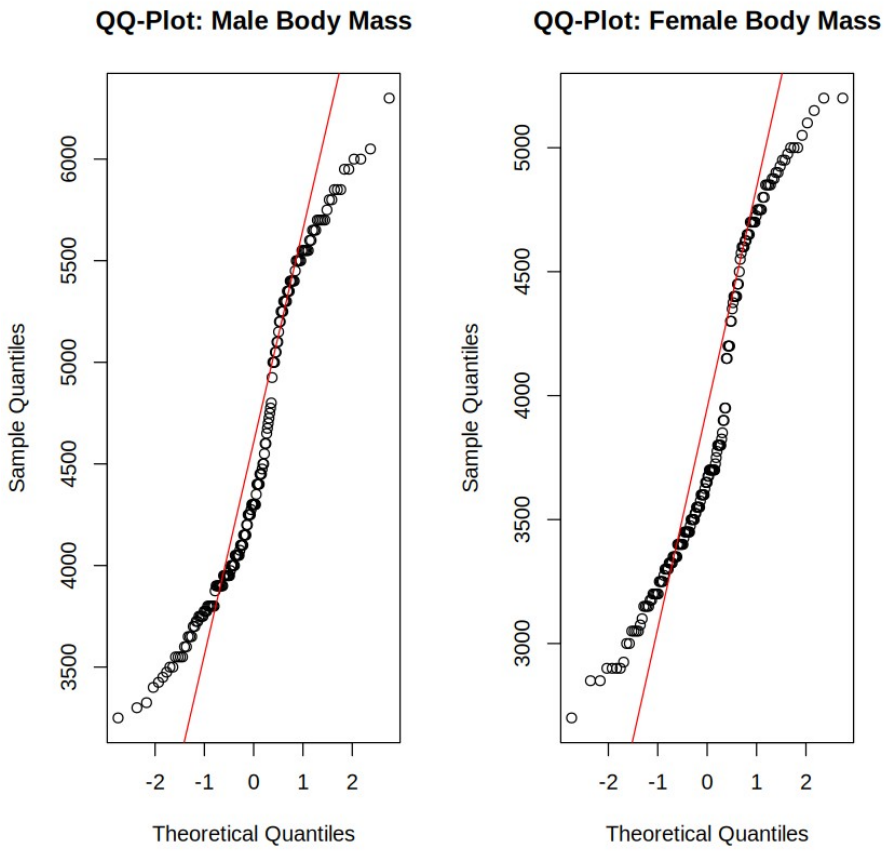


Figure 2.1: QQ-plots for male and female body mass

```
leveneTest(body_mass_g ~ sex, data = penguins)

      Df  F value  Pr(>F)
group   1 6.058617 0.01434821
      331       NA         NA

t_test_sex <- t.test(body_mass_g ~ sex, data = penguins, var.equal =
FALSE)
t_test_sex

       Welch Two Sample t-test

data:  body_mass_g by sex
t = -8.5545, df = 323.9, p-value = 4.794e-16
alternative hypothesis: true difference in means between group female
and group male is not equal to 0
95 percent confidence interval:
 -840.5783 -526.2453
sample estimates:
mean in group female   mean in group male
            3862.273             4545.685

cohen_d <- function(x, y) {
  nx <- length(x); ny <- length(y)
  pooled_sd <- sqrt(((nx - 1) * var(x) + (ny - 1) * var(y)) / (nx + ny
- 2))
  (mean(x) - mean(y)) / pooled_sd
}
d_sex <- cohen_d(male_mass, female_mass)
cat("Cohen's d (male vs female):", round(d_sex, 3), "\n")

Cohen's d (male vs female): 0.936
```

	``` cat("p-value =", format.pval(t_test_sex$p.value, digits = 4), "-> ", ifelse(t_test_sex$p.value < 0.05, "Reject H0: mean body mass differs significantly by sex.", "Fail to reject H0: no significant difference by sex."), "\n") cat("Effect size (|d| =", round(abs(d_sex), 2), ") indicates a", ifelse(abs(d_sex) < 0.2, "negligible", ifelse(abs(d_sex) < 0.5, "small", ifelse(abs(d_sex) < 0.8, "medium", "large"))), "effect.\n") p-value = 4.794e-16 -> Reject H0: mean body mass differs significantly by sex. Effect size (|d| = 0.94) indicates a large effect. ```
Output	``` N (after removing NA): 333 Overall mean body mass: 4207.06 g	Overall SD: 805.22 Species means (g): Adelie 3706.16	Chinstrap 3733.09	Gentoo 5092.44 Shapiro-Wilk p (male): 1.227e-07 Shapiro-Wilk p (female): 6.155e-08 Levene's test (sex) p: 0.01435 Welch t-test: t = -8.5545, df = 323.9, p = 4.794e-16 95% CI for mean difference (female - male): -840.58 to -526.25 g Cohen's d: 0.936 (large effect) ```
Conclusion	Both Shapiro-Wilk tests were significant ($p < 0.05$), indicating body mass is not perfectly normally distributed within either sex group, though the QQ-plots show only mild deviation, mainly in the tails. Levene's test indicated a small but significant difference in variances ($p = 0.014$), so a Welch t-test (which does not assume equal variances) was used. The Welch t-test found a highly significant difference in mean body mass between males and females ($p = 4.79e-16$), with males averaging roughly 683 g heavier than females. The 95% CI (-840.6 to -526.2 g) excludes zero, confirming the effect is real, and Cohen's $d = 0.94$ indicates a large effect size: sex is a strong predictor of penguin body mass in this population.

Experiment Number	**6**
Experiment Description	**Lab 2: Statistical Analysis of Palmer Penguins: ANOVA & Non-Parametric Tests** Topic: One-Way & Two-Way ANOVA, Post-Hoc Tests, Non-Parametric Tests | Dataset: Palmer Penguins Dataset **Problem Statement** Building on Experiment 5, this lab investigates whether body mass and flipper length differ significantly across the three penguin species, and whether species and sex jointly influence body mass, using one-way ANOVA, Tukey HSD post-hoc comparisons, the Kruskal-Wallis test and two-way ANOVA.

Process	**Task 3: One-Way ANOVA (Body Mass Across Species)**
	`by(penguins$body_mass_g, penguins$species, shapiro.test)`
	`penguins$species: Adelie`
	`Shapiro-Wilk normality test`
	`data: dd[x, ]`
	`W = 0.98116, p-value = 0.04232`
	`------------------------------------------------------------`
	`penguins$species: Chinstrap`
	`Shapiro-Wilk normality test`
	`data: dd[x, ]`
	`W = 0.98449, p-value = 0.5605`
	`------------------------------------------------------------`
	`penguins$species: Gentoo`
	`Shapiro-Wilk normality test`
	`data: dd[x, ]`
	`W = 0.98606, p-value = 0.2605`

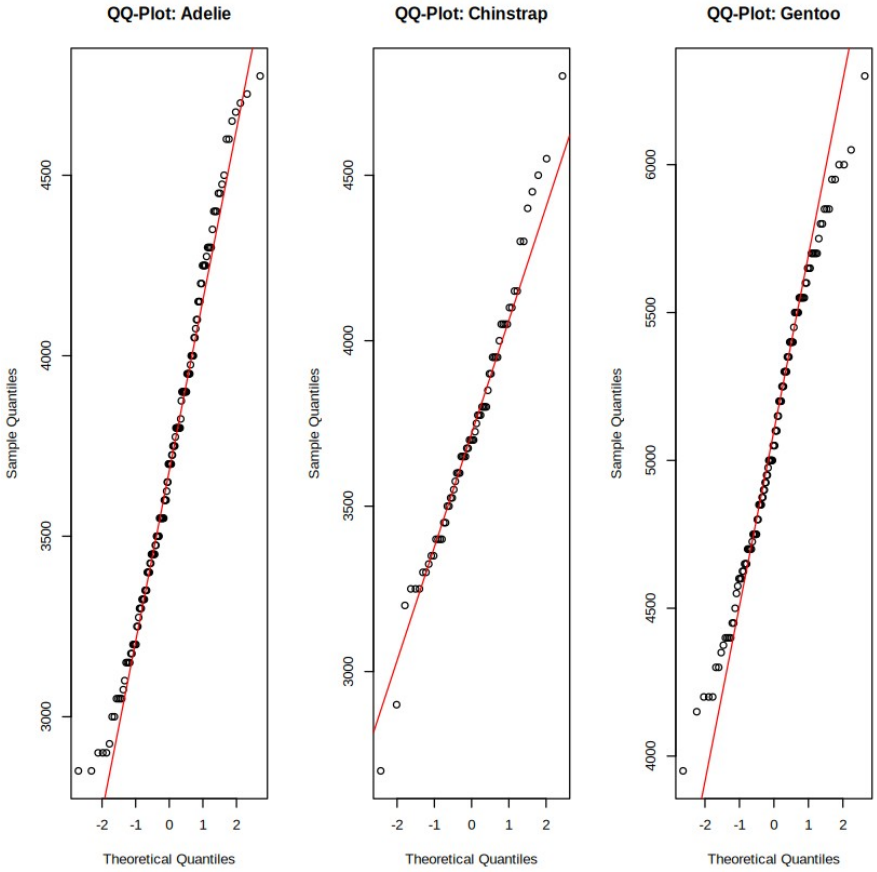


Figure 3.1: QQ-plots of body mass for each species

```
leveneTest(body_mass_g ~ species, data = penguins)

      Df  F value  Pr(>F)
group   2 5.134899 0.006367054
      330       NA          NA

anova_species <- aov(body_mass_g ~ species, data = penguins)
summary(anova_species)

             Df    Sum Sq  Mean Sq F value Pr(>F)
species       2 145190219 72595110   341.9 <2e-16 ***
Residuals   330  70069447   212332
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

F( 2 , 330 ) = 341.89 , p = < 2.2e-16

if (p_value < 0.05) {
  tukey_species <- TukeyHSD(anova_species)
  print(tukey_species)
  plot(tukey_species)
}

Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = body_mass_g ~ species, data = penguins)

$species
                        diff       lwr       upr    p adj
Chinstrap-Adelie   26.92385 -132.3528  186.2005 0.916431
Gentoo-Adelie    1386.27259 1252.2897 1520.2554 0.000000
```

```
Gentoo-Chinstrap 1359.34874 1194.4304 1524.2671 0.000000
```

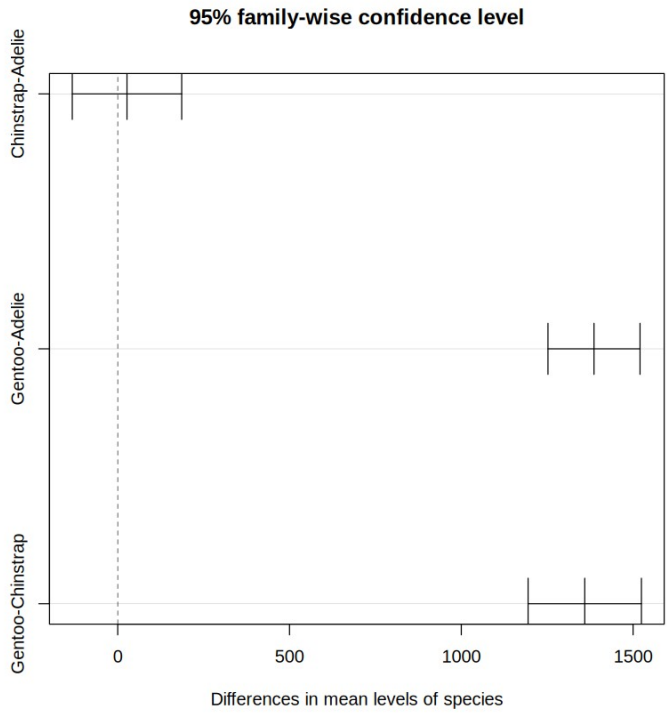


Figure 3.2: Tukey HSD 95% family-wise confidence intervals

Task 4: Non-Parametric Analysis (Kruskal-Wallis)

```
kruskal_species <- kruskal.test(body_mass_g ~ species, data =
penguins)
kruskal_species

       Kruskal-Wallis rank sum test

data:  body_mass_g by species
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16

cat("ANOVA p-value          :", format.pval(p_value, digits = 4), "\
n")
cat("Kruskal-Wallis p-value :", format.pval(kruskal_species$p.value,
digits = 4), "\n")
cat(ifelse((p_value < 0.05) == (kruskal_species$p.value < 0.05),
           "Both tests lead to the SAME overall conclusion.",
           "The two tests DISAGREE in their conclusion."), "\n")

ANOVA p-value          : < 2.2e-16
Kruskal-Wallis p-value : < 2.2e-16
Both tests lead to the SAME overall conclusion.
```

Task 5: Two-Way ANOVA (Species x Sex on Body Mass)

```
twoway_anova <- aov(body_mass_g ~ species * sex, data = penguins)
summary(twoway_anova)

             Df    Sum Sq  Mean Sq F value   Pr(>F)
species       2 145190219 72595110 758.358  < 2e-16 ***
sex           1  37090262 37090262 387.460  < 2e-16 ***
species:sex   2   1676557   838278   8.757 0.000197 ***
Residuals   327  31302628    95727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

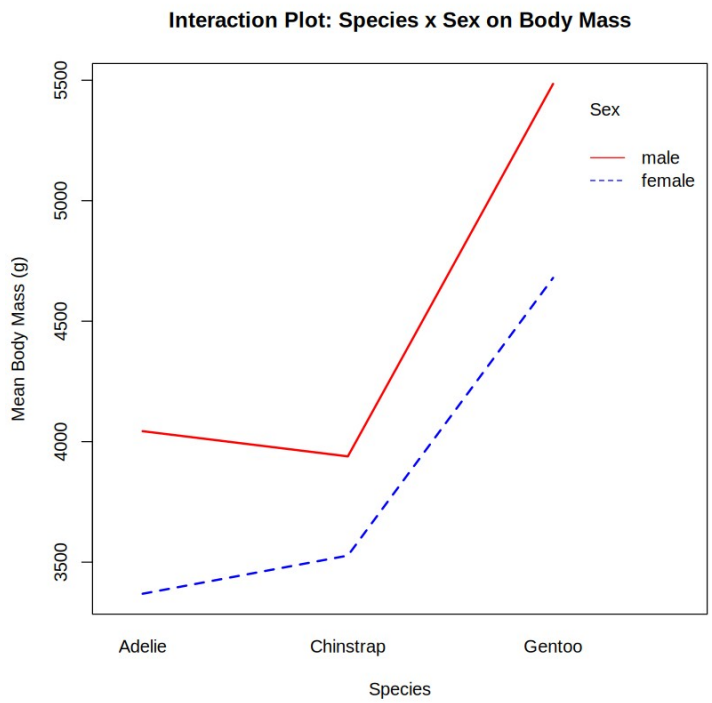


Figure 5.1: Interaction plot, species x sex on body mass

Task 6: Flipper Length Analysis

```
flipper_desc <- penguins %>%
  group_by(species) %>%
  summarise(
    Mean = mean(flipper_length_mm),
    Median = median(flipper_length_mm),
    SD = sd(flipper_length_mm),
    Skewness = skewness(flipper_length_mm),
    Kurtosis = kurtosis(flipper_length_mm)
  )
flipper_desc

  species   Mean     Median SD       Skewness     Kurtosis
1 Adelie    190.1027 190    6.521825  0.079578511 3.327738
2 Chinstrap 195.8235 196    7.131894 -0.009262203 2.956063
3 Gentoo    217.2353 216    6.585431  0.364085778 2.322009

by(penguins$flipper_length_mm, penguins$species, shapiro.test)

penguins$species: Adelie

       Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.9934, p-value = 0.7427

------------------------------------------------------------
penguins$species: Chinstrap

       Shapiro-Wilk normality test

data:  dd[x, ]
W = 0.98891, p-value = 0.8106

------------------------------------------------------------
penguins$species: Gentoo

       Shapiro-Wilk normality test
```

```
data:  dd[x, ]
W = 0.96148, p-value = 0.00176

leveneTest(flipper_length_mm ~ species, data = penguins)

      Df  F value   Pr(>F)
group   2 0.4427865 0.6426253
      330        NA        NA

anova_flipper <- aov(flipper_length_mm ~ species, data = penguins)
summary(anova_flipper)

             Df Sum Sq Mean Sq F value Pr(>F)
species       2  50526   25263   567.4 <2e-16 ***
Residuals   330  14693      45
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

flipper_summary <- summary(anova_flipper)[[1]]
f_flipper <- flipper_summary["species", "F value"]
p_flipper <- flipper_summary["species", "Pr(>F)"]
cat("Flipper length ANOVA: F =", round(f_flipper, 2),
    ", p =", format.pval(p_flipper, digits = 4), "\n")

if (p_flipper < 0.05) {
  tukey_flipper <- TukeyHSD(anova_flipper)
  print(tukey_flipper)
}

Flipper length ANOVA: F = 567.41 , p = < 2.2e-16
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = flipper_length_mm ~ species, data = penguins)

$species
                     diff       lwr       upr p adj
Chinstrap-Adelie  5.72079  3.414364  8.027215     0
Gentoo-Adelie    27.13255 25.192399 29.072709     0
Gentoo-Chinstrap 21.41176 19.023644 23.799885     0

kruskal_flipper <- kruskal.test(flipper_length_mm ~ species, data =
penguins)
kruskal_flipper

       Kruskal-Wallis rank sum test

data:  flipper_length_mm by species
Kruskal-Wallis chi-squared = 237.35, df = 2, p-value < 2.2e-16
```

Task 7: Visualizations

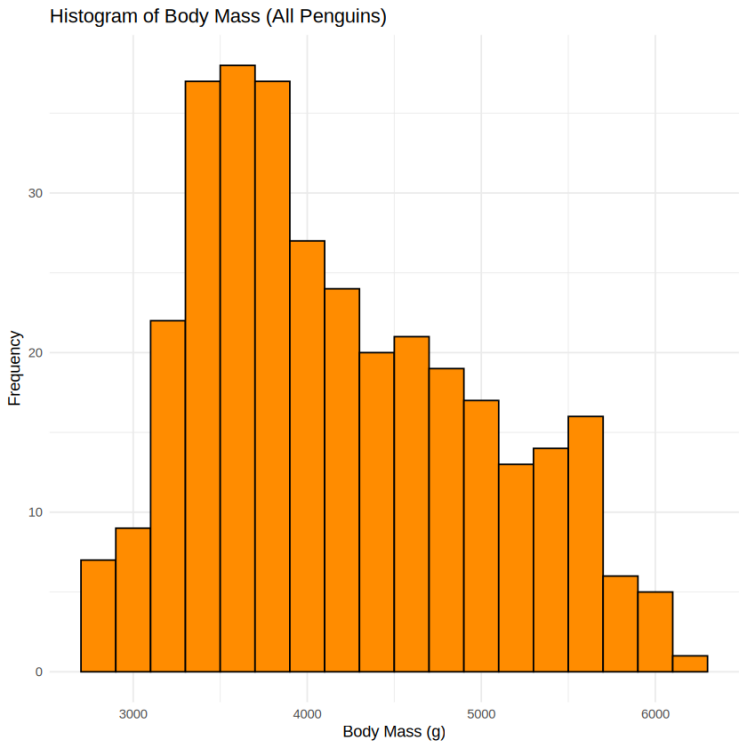


Figure 7.1: Histogram of body mass (all penguins)

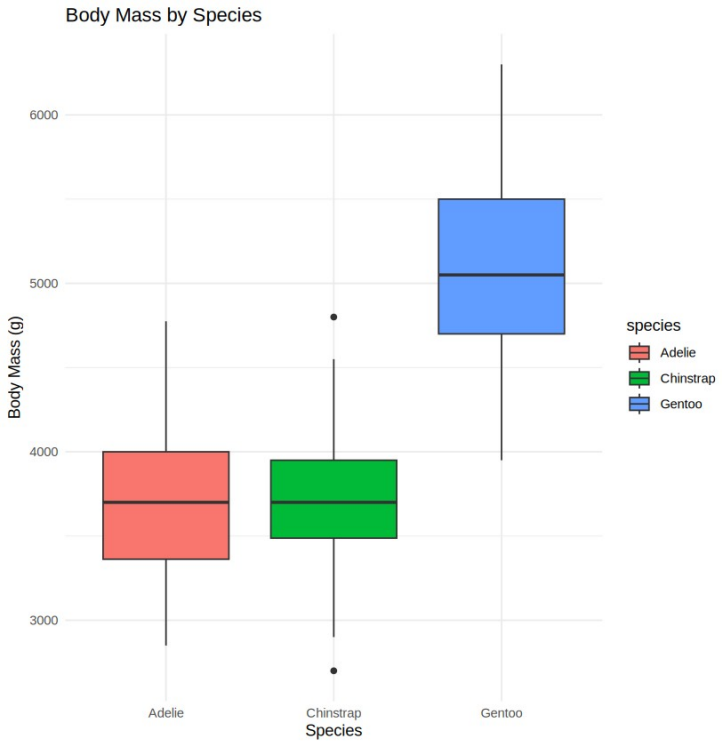


Figure 7.2: Body mass by species

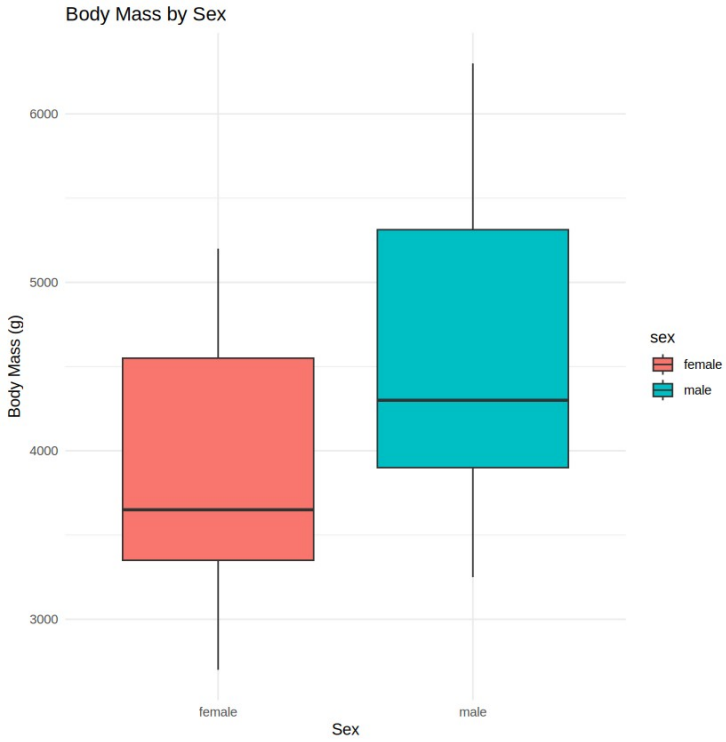


Figure 7.3: Body mass by sex

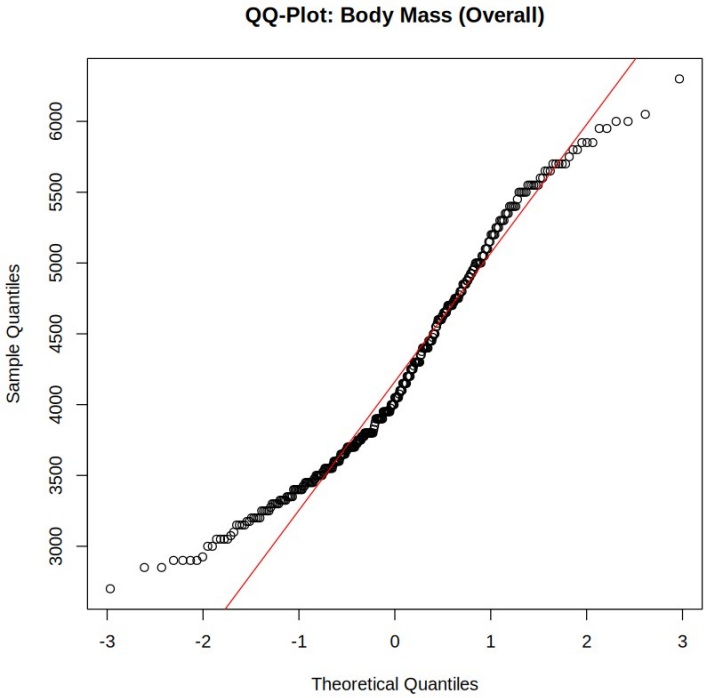


Figure 7.4: QQ-plot of body mass (overall)

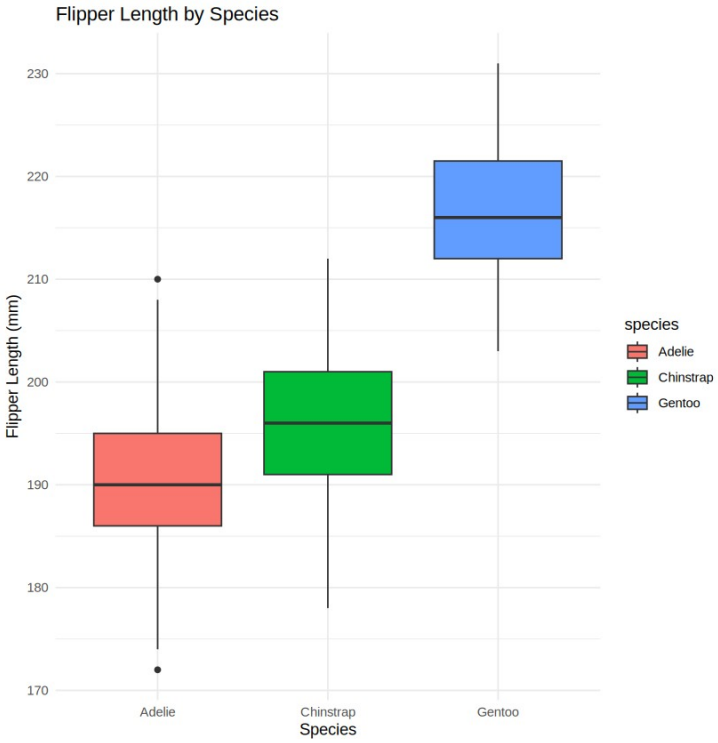


Figure 7.5: Flipper length by species

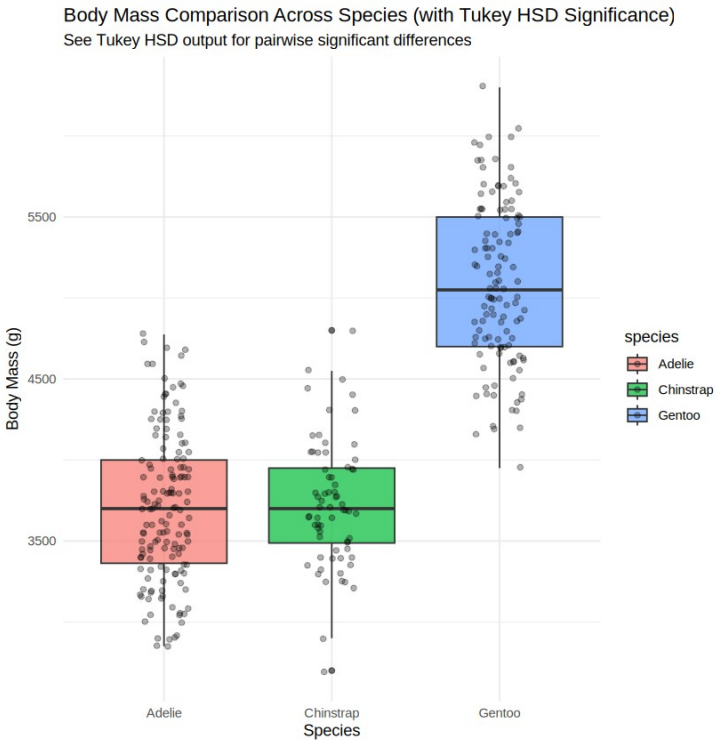


Figure 7.6: Body mass comparison across species with individual data points

Output

```
One-way ANOVA (body mass ~ species): F(2,330) = 341.89, p < 2.2e-16
Tukey HSD: Gentoo differs significantly from both Adelie and Chinstrap
(p < 0.001);
           Chinstrap vs Adelie not significant (p = 0.916)
Kruskal-Wallis (body mass): chi-sq = 212.09, df = 2, p < 2.2e-16
Two-way ANOVA: species p < 2.2e-16 | sex p < 2.2e-16 | species:sex p =
```

	1.973e-04 Flipper length ANOVA: F(2,330) = 567.41, p < 2.2e-16 Flipper length Kruskal-Wallis: chi-sq = 237.35, df = 2, p < 2.2e-16
Conclusion	The one-way ANOVA confirmed that mean body mass differs significantly across species (F = 341.9, p < 2.2e-16), and Tukey HSD showed this is driven entirely by Gentoo penguins, which are significantly heavier than both Adelie and Chinstrap (p < 0.001 for both), while Adelie and Chinstrap do not differ significantly from each other (p = 0.92). The Kruskal-Wallis test reached the same conclusion, so the result is robust even though Levene's test flagged mild heteroscedasticity across species. The two-way ANOVA showed that both species and sex significantly affect body mass, and a significant species x sex interaction (p < 0.001) indicates the male-female weight gap is not identical across species. The flipper-length analysis mirrors the body-mass findings: Gentoo penguins have significantly longer flippers than Adelie and Chinstrap (Tukey HSD, p < 0.001 for both), and Chinstrap also differ significantly from Adelie, reinforcing that Gentoo penguins are structurally the largest of the three species.